

```

1 pm_filter <- function(n_sig, n_fil, freq, type, d1=.05, d2=.07) {
2   c0 <- 2 * freq / n_sig
3
4   if(type == "stop") {
5     v <- c(1,1,0,0,1,1)
6   } else if(type == "pass") {
7     v <- c(0,0,1,1,0,0)
8   }
9
10  fir <- remez(n = n_fil, f=c(0,c0-d2,c0-d1,c0+d1,c0+d2,1), a = v)
11  freq <- freq*(fir, n=n_fil)
12  return(list(fir = fir, freq = freq))
13 }
14
15 apply_filter_to_sig <- function(fir, sig) {
16   y <- signal::filter(as.vector(fir), 1, x = sig)
17 }
18
19 # signal length
20 n <- 100000
21
22 # filter length
23 n_fil <- 800
24
25 # the frequency to be filtered out
26 F <- 5000
27 F0 <- 2 * F / n
28
29 sig <- sin_sig_incr_freq(n, from = 0, to = 10000)
30
31 filter <- pm_filter(
32   n_sig = n, n_fil = n_fil, freq = F, type = "stop", d1 = F0*.05, d2 = F0*.1)
33
34 sig2 <- apply_filter_to_sig(filter$fir, sig)

```

Fig. 2: Sample R code (Courtesy: www.joyofdata.de)

and statistical models like regression and tree model are prepared. These techniques allow academicians and researchers around the world to develop latest methods in statistics, machine learning and predictive modelling. There are thousands of packages in every domain that extend the capabilities of R to adapt to various applications and their number increases day by day.

Faster yet reliable results. With the ability to mix and match models to yield better results, a normal R programmer can yield faster, yet accurate, results. The code can be automated and thus be reproduced, nourishing greater research.

Challenges faced while coding in R

Engineers who are new to R face certain challenges when they are introduced to R for the first time.

Mastering the language. R is not an easy-to-learn language. The

documentation style could have been more user-friendly so that the language would have more user base.

Memory and speed issues. R software is designed to be more generic in nature and programmers code it to suite their applications. Till date there are over 2000 applications developed extending the functionalities of R. Obviously, R is not the best for certain applications. There are memory and speed issues when extending the code for certain applications.

Syntax of the program. For a professional programmer, syntax of the script might look somewhat untidy when compared to other languages like Python.

How a communication engineer can make use of R

As discussed earlier, R is designed to be a generic tool and not an electronic design automation tool.

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